

~~m~~ dataset

Load (x) (N)

Extension (y) (mm)

5

1.2

10

2.5

15

3.7

20

4.9

25

6.1

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$n = 5$$

$$\sum xy = 337 \quad \sum x = 75 \quad \sum y = 18.4 \quad \sum x^2 = 1375$$
$$(\sum x)^2 = 5625$$

$$\therefore m = \cancel{5} \times 0.244$$

$$c = \frac{\sum y - m \sum x}{n} = \frac{18.4 - 0.244 \times 75}{5}$$
$$= 0.02$$

$$y = 0.244x + 0.02$$

For load of 55N, extension can be found by
putting $x = 55\text{N} \Rightarrow y = (\text{extension for } 55\text{N})$
 $= 13.44\text{mm}$